Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination 2018
Year 6

H1 CHEMISTRY 8873/02

Paper 2 Structured Questions

13 September 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Answer Paper
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use								
P1	**P2 Section A**				**P2 Section B**		**Total**	**%**
	Q1	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**		
30	13	9	20	18	20	20	110	

Section A

Answer **all** the questions in this section in the spaces provided.

1 **(a)** Group 1 elements are strong reducing agents. State and explain the trend of reducing power of Group 1 elements down the Group.

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…………………………………………………………………………………………………

…………………………………………………………………………………………………[3]

(b) Group 1 elements and its compounds have many uses. Sodium hydrogencarbonate ($NaHCO_3$) is often used to treat ant stings which contains methanoic acid (HCOOH), a weak acid.

Write an equation for the dissociation of methanoic acid in water. Indicate which species are the acid, the base and their conjugate pairs in the reaction.

[2]

(c) When the ant bites, it injects a solution containing 50 % by volume of unionised methanoic acid. A typical ant may inject around 6.0×10^{-3} cm^3 of this solution.

(i) Given that the density of methanoic acid is 1.2 g cm^{-3}, what is the amount (in moles) of methanoic acid that a typical ant injects?

[1]

As soon as the methanoic acid is injected, it dissolves in water in the body to produce a solution of methanoic acid with pH 2.43.

(ii) Assuming that it dissolves fully in 1.0 cm^3 of water in the body, calculate the concentration of the methanoic acid solution that is formed initially.
You may ignore the volume of methanoic acid injected in this calculation.

[1]

(iii) Calculate the percentage of methanoic acid molecules which have dissociated in 1.0 cm^3 of water.

[2]

(d) A student was given an unlabelled bottle and was told that it contained pure sample of one of the following three compounds.

- methanol
- methanal
- propanone

(i) State the reagent used for a chemical test that could show that the sample can be either methanal and propanone but **not** methanol. Describe what would be observed.

Reagent: ……………………………………………………………………………[1]

Observations: ……………………………………………………………...……[1]

(ii) Describe one chemical test that could distinguish between methanal and propanone.

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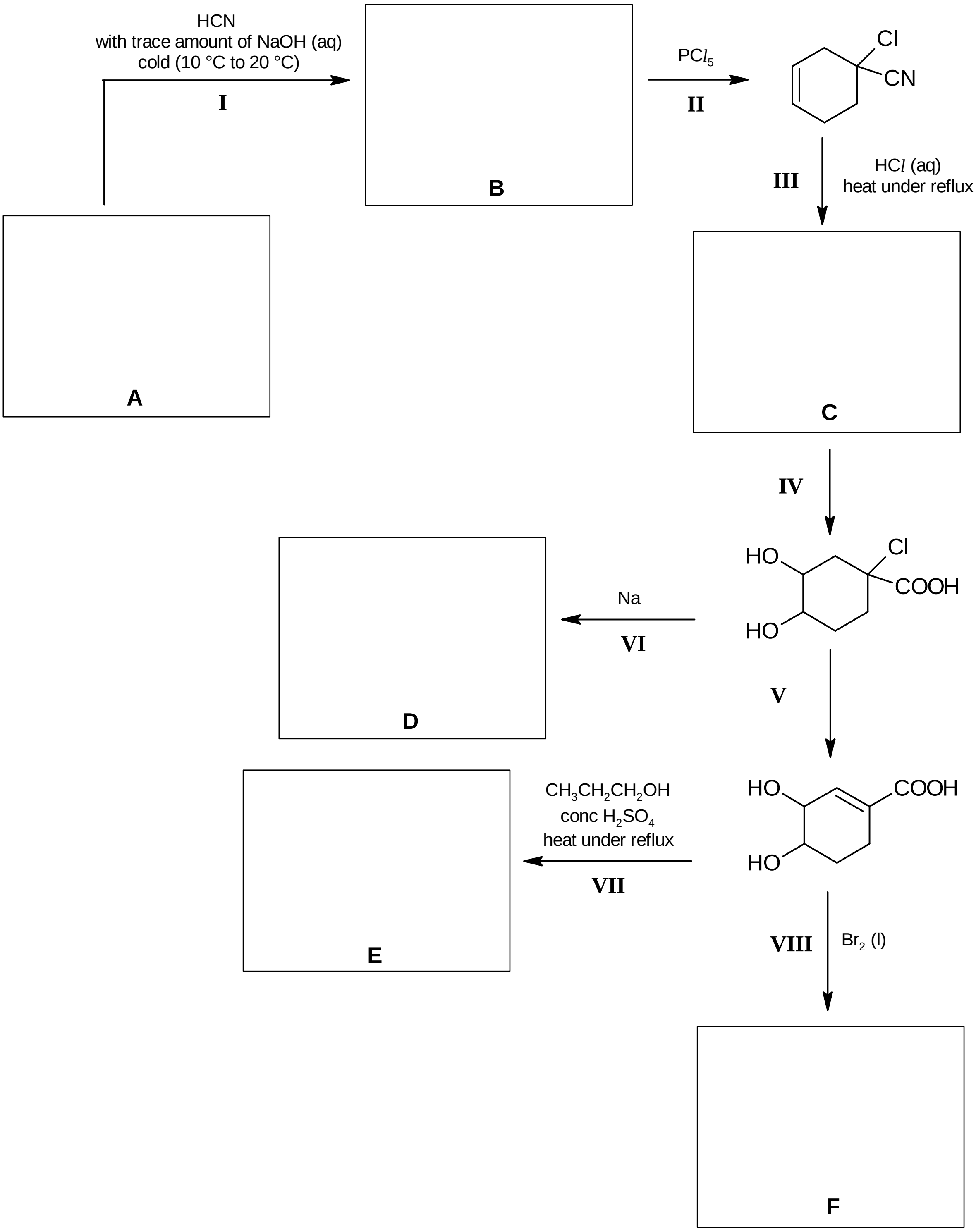
……………………………………………………………………………………….

……………………………………………………………………………………[2]

[Total: 13]

2 A sequence of reactions starting from compound **A** is shown below.

(a) Draw the structures of compounds **A**, **B**, **C**, **D**, **E** and **F** in the boxes below.



[6]

(b) For the reaction scheme shown above, state

(i) the type of reaction occurring in reaction **I**.

……………………………………………………………………………………………[1]

(ii) the reagents and conditions for reaction **V**.

……………………………………………………………………………………………[1]

(c) The alcohol $CH_3CH_2CH_2OH$, used as the reagent in reaction **VII** above, can be converted into $CH_3CH{=}CH_2$. How may this conversion be achieved in a laboratory?

……………………………………………………………………………………………………

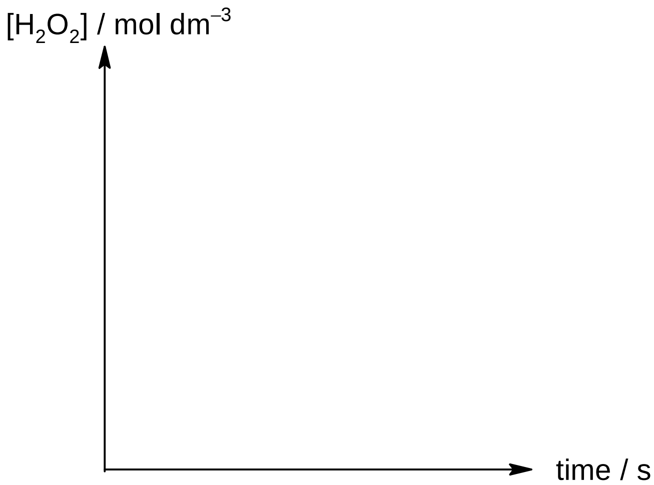
……………………………………………………………………………………………………[1]

[Total: 9]

3 **(a)** Pure hydrogen peroxide, H_2O_2, was long believed to be unstable. Its decomposition follows a first order reaction.

$$H_2O_2 \rightarrow H_2O + \tfrac{1}{2}O_2$$

(i) Sketch a graph of $[H_2O_2]$ against time to show that the reaction is first order with respect to H_2O_2.



[2]

(ii) As H_2O_2 decomposes slowly at room temperature, catalysts such as platinum metal are often added to lower the activation energy to increase the rate of reaction.

With reference to the information provided below, sketch the energy profile diagram showing the catalysed reaction only.

enthalpy change of the decomposition	–196 kJ mol^{-1}
activation energy (without catalyst)	+75 kJ mol^{-1}
activation energy (with catalyst)	+49 kJ mol^{-1}

[2]

(b) In the presence of UV light, H_2O_2 decomposes to form hydroxyl free radicals, •OH.

(i) Draw the 'dot–and–cross' diagram for H_2O_2.

[1]

(ii) Using relevant bond energy values from the *Data Booklet*, suggest the relative rate of the formation of •C*l* from chlorine gas as compared that of •OH from hydrogen peroxide.

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…………………………………………………………………………………………[2]

(iii) Propylamine, $CH_3CH_2CH_2NH_2$, may be formed via the following reaction pathway involving free radical substitution in the first step.

$$CH_3CH_2CH_3 \xrightarrow[Cl_2,\ uv]{\textbf{Step I}} CH_3CH_2CH_2Cl \xrightarrow{\textbf{Step II}} CH_3CH_2CH_2NH_2$$

Name the type of reaction occurring in **Step II**. State the reagent and conditions required for this step.

Type of reaction: ……………………………………………………………………

Reagent: ……………………………………………………………………………

Conditions: …………………………………………………………………………

…………………………………………………………………………………………

[2]

(iv) As the yield from the reaction in **(b)(iii)** is low, propose a 3-step synthetic route to produce propylamine from ethene instead. Show clearly the reagents and conditions as well as the intermediates involved.

[3]

(c) The kinetics of the reaction between hydrogen peroxide and iodide ions in acidic solution was studied.

$$H_2O_2 + 2I^- + 2H^+ \rightarrow 2H_2O + I_2$$

Four separate experiments were carried out to determine the relative rates by varying the concentrations of the reactants. The results obtained are given in the table below.

Expt	$[H_2O_2]$ / mol dm^{-3}	$[I^-]$ / mol dm^{-3}	$[H^+]$ / mol dm^{-3}	relative rate
1	0.03	0.03	0.03	1.0
2	0.05	0.03	0.03	1.6
3	0.05	0.01	0.06	0.53
4	0.03	0.01	0.03	0.33

(i) With reference to the overall equation given above, write the oxidation and reduction half-equations.

Oxidation half-equation: ……………………………………………………………[1]

Reduction half-equation: ……………………………………………………………[1]

(ii) Use the data to deduce the orders of reaction with respect to H_2O_2, I^- and H^+. Show your working clearly.

[3]

(iii) Hence, write the rate equation and state the units of the rate constant.

[2]

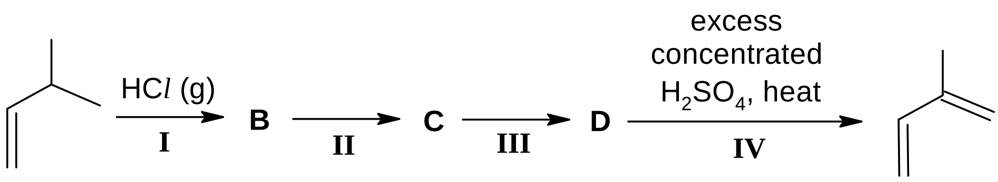
(iv) Unreacted iodide ions may be easily separated from the reaction mixture by adding silver nitrate solution, followed by filtration. State the identity and colour of the precipitate formed.

……………………………………………………………………………………….[1]

[Total : 20]

4 Isoprene, **E**, is an organic compound that could be used to synthesise limonene, which is commonly used in fragrances.

(a) **E** can be synthesised from 3–methylbut–1–ene, **A**, in a 4–step process as follows.



A **E**

(i) **B** is a major product of step **I**.

Draw the structures of compounds **B**, **C** and **D**.

[3]

B **C** **D**

(ii) Hydrogen chloride, the reagent in step **I**, is commonly used in organic synthesis.

State and explain how the thermal stabilities of hydrogen halides varies down the Group.

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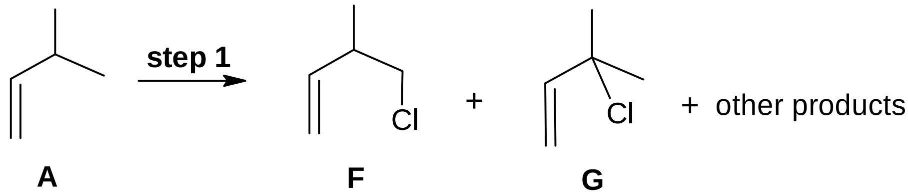
(b) The following reaction shows an alternative route to form an intermediate for the synthesis of isoprene.



(i) State the reagent and condition of **step 1** which will result in formation of compounds **F** and **G**.

Reagent: …………………….…………….....

Condition: ……………………………….…..

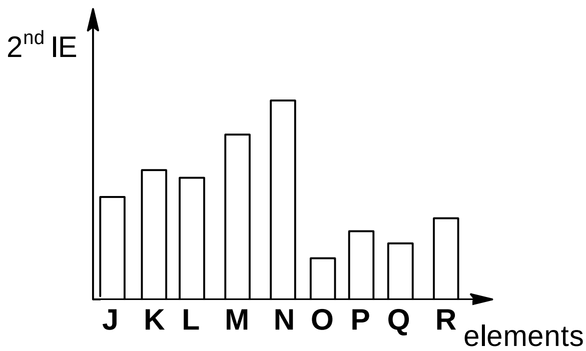
[1]

(ii) Predict the ratio in which **F** and **G** will be formed.

compound	**F**	**G**
ratio		

[1]

(c) The bar chart below shows the second ionisation energy (2nd IE) of nine consecutive elements (**J** to **R**) in Periods 2 and 3 of the Periodic Table.



(i) Write an equation for the second ionisation energy of element **J**.

……………………………………………………………………………………….……[1]

(ii) Identify element **N**.

Element **N** is ……………………………………..….[1]

(iii) Using your answers in **(c)(i)**, **(c)(ii)** and the electronic configurations of the species involved, explain the significantly higher 2nd IE of **N** compared to **O**.

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……………………………………………………………………………………………

……………………………………………………………………………………………[2]

(d) The Periodic Table shows the element, Helium, placed at the top of Group 18.

(i) Suggest why the element Helium could be placed at the top of Group 2.

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……………………………………………………………………………………………[1]

(ii) Suggest why the element Helium is not placed at the top of Group 2, by comparing **one** physical property. Explain your answer.

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……………………………………………………………………………………………

……………………………………………………………………………………………[2]

(e) Al_2O_3, SiO_2 and P_4O_{10} are oxides of Period 3 elements.

(i) Explain why both Al_2O_3 and SiO_2 are insoluble in water.

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……………………………………………………………………………………………[2]

(ii) Explain, with the aid of an equation, the reaction of P_4O_{10} with water.

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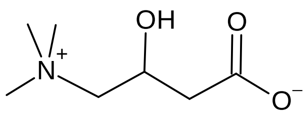
……………………………………………………………………………………………[2]

[Total: 18]

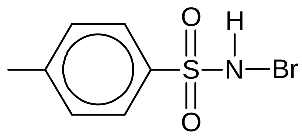
Section B

Answer **one** question from this section, in the spaces provided.

5 Levocarnitine is a quaternary ammonium compound involved in metabolism in most mammals.



levocarnitine



bromamine-T

(a) A kinetic study on the $RuCl_3$ catalysed reaction between levocarnitine and bromamine-T was carried out in aqueous hydrochloric acid.

(i) The following equilibrium exists for $RuCl_3$ in aqueous hydrochloric acid.

$$[RuCl_5(H_2O)]^{2-} + Cl^- \rightleftharpoons [RuCl_6]^{3-} + H_2O$$

Explain why $[RuCl_6]^{3-}$ is likely to be the reactive species instead of $[RuCl_5(H_2O)]^{2-}$ in this study.

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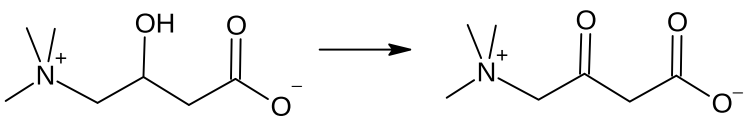
……………………………………………………………………………………………[1]

(ii) Levocarnitine is converted into the following product by bromamine-T.



levocarnitine

State the type of reaction that levocarnitine had undergone and explain your answer in terms of changes in oxidation number.

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……………………………………………………………………………………………[2]

(b) A series of experiments were carried out at different temperatures under first order conditions with respect to bromamine-T.

The value of the observed rate constant, k, for the catalysed reaction was determined at each temperature and the results are summarised in the table below.

k / 10^4 s^{-1}	temperature, T / K
1.82	293
3.00	303
4.62	313
7.30	323

The activation energy, E_a, and the pre-exponential factor, A, which is a constant, for the reaction can be determined from the equation.

$$k = Ae^{\frac{-E_a}{RT}}$$

R is the molar gas constant.
T is the reaction temperature in kelvin.
k is the observed rate constant at a chosen temperature.

(i) Calculate the values of ln k and $\frac{1}{T}$ for each of the experiments above.

k / 10^4 s^{-1}	ln k	temperature, T / K	$\frac{1}{T}$ / K^{-1}
1.82		293	
3.00		303	
4.62		313	
7.30		323	

[2]

(ii) Hence plot a graph of ln k against $\frac{1}{T}$ and determine E_a from the gradient of the best-fit line which is $\frac{-E_a}{R}$.

Gradient ……………………………………………..

E_a ……………………………………………..

[4]

(iii) State the effect of temperature on the rate of the reaction and explain your answer with the aid of a Boltzmann distribution curve.

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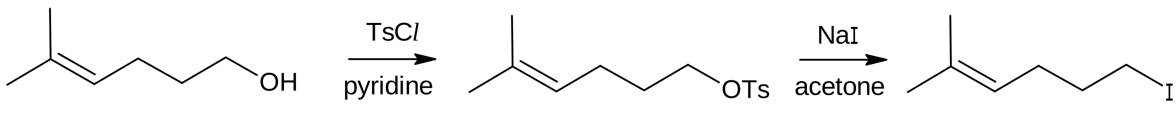
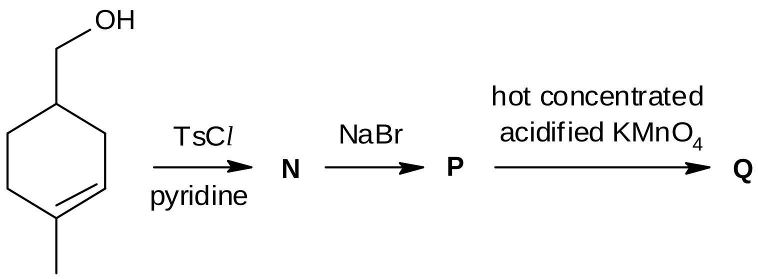
……………………………………………………………………………………[4]

Tosyl chlorides (TsC*l*) are often used to convert alcohols into alkyl tosylates which undergo substitution reactions. An example of this application is given below.



alkyl tosylate

(c) Consider the reaction scheme below involving an alkyl tosylate **N**.



M

(i) State the functional group(s) present in the starting material, **M**.

……………………………………………………………………………………[2]

(ii) Describe what will be observed when **M** is heated with acidified potassium dichromate(VI) and draw the structure of the organic product formed.

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……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………[2]

(iii) Draw the structures of the compounds **N**, **P** and **Q**.

N	**P**	**Q**

[3]

[Total: 20]

6 Methanoic acid is the simplest carboxylic acid and is an important intermediate in chemical synthesis. It occurs naturally and is found notably in ants.

(a) **(i)** Methanoic acid is a *weak acid*. What do you understand by the term in *italics*?

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……………………………………………………………………………………………[2]

(ii) The acid dissociation constant, K_a, for methanoic acid is 1.6×10^{-4} mol dm^{-3}. Calculate the pH of a 0.100 mol dm^{-3} solution of methanoic acid, given that $[H^+] = \sqrt{K_a \times [HCOOH]}$.

[2]

(iii) A solution comprising methanoic acid and sodium methanoate can function as a buffer solution. With the aid of a balanced equation, describe how this buffer solution can resist pH changes when a small amount of sulfuric acid is added.

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……………………………………………………………………………………………[2]

(b) The energy contents of methanoic acid can be determined by means of calorimetric experiments. These experiments are usually carried out using polystyrene cup in a normal school laboratory. The standard enthalpy change of neutralisation can also be determined.

(i) Define what is meant by the *standard enthalpy change of neutralisation*.

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……………………………………………………………………………………………[1]

(ii) Write a balanced chemical equation for the neutralisation of methanoic acid with potassium hydroxide.

…………………………………………………………………………………………[1]

(iii) How would you expect the enthalpy change of neutralisation in **(b)(ii)** to compare with the enthalpy change of neutralisation of nitric acid with potassium hydroxide? Explain your answer.

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……………………………………………………………………………………………

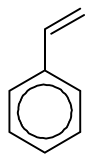
……………………………………………………………………………………………

…………………………………………………………………………………………[2]

(iv) Suggest a suitable indicator for the titration between methanoic acid and potassium hydroxide.

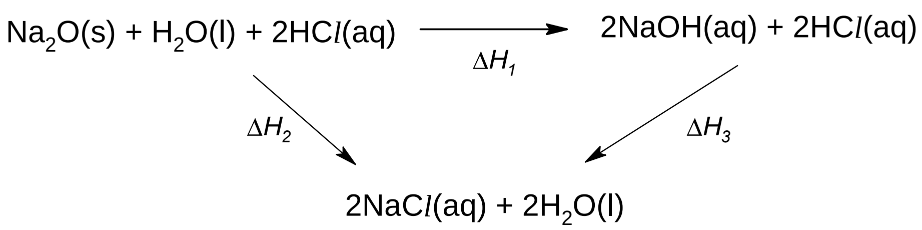
…………………………………………………………………………………………[1]

(v) Draw the structure of polystyrene (with at least three repeating units) given that the monomer styrene has the structure as shown below.



[1]

(c) Enthalpy change of reaction 1, ΔH_1, can be determined using enthalpy change of reaction 2, ΔH_2, and enthalpy change of reaction 3, ΔH_3, in the energy cycle below.



When 6.2 g of $Na_2O(s)$ is dissolved in 250 cm^3 of 1.0 mol dm^{-3} $HCl(aq)$, the temperature of the solution rose by 17 ºC.

(i) Using the cycle above, calculate ΔH_2.

[3]

(ii) The enthalpy change of neutralisation between $NaOH(aq)$ and $HCl(aq)$ is known to be –57.3 kJ mol^{-1}. Calculate ΔH_1.

[1]

(d) The German company BASF is a world leader in the production of methanoic acid. It has an efficient process of producing methanoic acid from methyl methanoate. The process is described below.

Step 1:
Methyl methanoate is mixed with water and allowed to react to produce methanol and methanoic acid.

Step 2:
Methanol is reacted with carbon monoxide and a catalyst to produce methanoic acid.

(i) State the type of reaction occurring in Step 1.

……………………………………………………………………………………………[1]

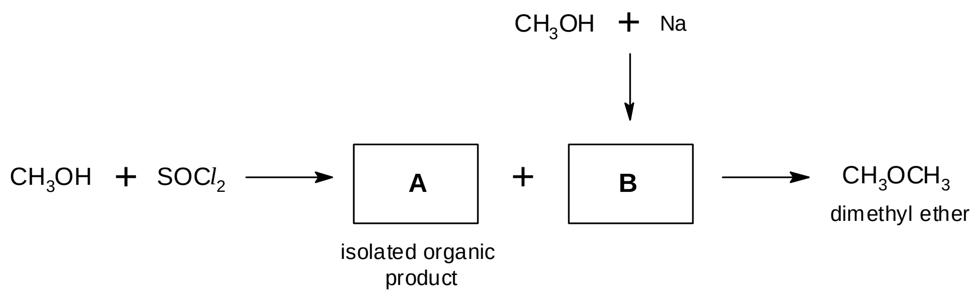
(ii) Suggest why a catalyst is used in Step 2.

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………[1]

(iii) Methanol produced using the process can also be used to produce dimethyl ether which is an alternative fuel to diesel.



Draw the displayed formulae of compounds **A** and **B**.

A

B

[2]

[Total: 20]

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